

The Guess and Check Method to Evaluate Integrals

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I Introduction

In an ordinary differential equations class, we need to be able to evaluate integrals very quickly, as they are often only one step of the solution to problem. Luckily, there are a few methods we can use to do this. Let $f(x)$ be a function that is Riemann integrable. Then, let the anti-derivative of $f(x)$ can be denoted by $F_p(x)$. Below are anti-derivatives to common functions.

$f(x)$	k	x^r	$\sin(x)$	$\cos(x)$	e^x	$\frac{1}{x}$	$\frac{1}{x^2 + a^2}$	$\frac{1}{\sqrt{a^2 - x^2}}$
$F_p(x)$	kx	$\frac{x^{r+1}}{r+1}$	$-\cos(x)$	$\sin(x)$	e^x	$\ln x $	$\frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right)$	$\sin^{-1}\left(\frac{x}{a}\right)$

II U-Substitution

For u-substitution, our goal is to guess an easy integral and then multiply it by a constant to get our final solution. For example, consider

$$\int \frac{1}{(5x-1)^{1/3}} dx$$

We know that u should equal $5x-1$. So, we will have

$$\int \frac{1}{u^{1/3}} \approx u^{2/3}$$

We know that the anti-derivative is some fraction times $u^{2/3}$ so that is our “guess”. Subbing back in, and taking the derivative to get our unknown fraction

$$(5x-1)^{2/3} \implies \frac{d}{dx}(5x-1)^{2/3} = \frac{2}{3}(5x-1)^{-1/3} \cdot 5 = \frac{10}{3}(5x-1)^{-1/3}$$

Finally we just multiply by the reciprocal of $\frac{10}{3}$ (since that is the constant we are off by) to arrive at our final answer

$$\frac{3}{10}(5x-1)^{2/3} + C$$

[†]Definitions and example problems from MATH 32044: Ordinary Differential Equations instructed by Dr. Todd Neal and *Lectures on Differential Equations* by Phillip Korman. Kent State University.

III Integration By Parts

For integration by parts, our goal is to integrate one of the functions, combine it with the other function, take the derivative, then subtract/add the anti derivative. For example, consider

$$\int 5xe^{3x}dx$$

First, integrate e^{3x} and multiply it by our other function, $5x$.

$$\int e^{3x}dx = \frac{e^{3x}}{3} \implies 5x \frac{e^{3x}}{3}$$

Then, take the derivative of our new product

$$\frac{d}{dx} 5x \frac{e^{3x}}{3} = 5xe^{3x} + \frac{5e^{3x}}{3}$$

Then, we will subtract the anti derivative of $\frac{5e^{3x}}{3}$ from our product we found by integrating e^{3x} (which is $5x \frac{e^{3x}}{3}$)

$$\int \frac{5e^{3x}}{3}dx = \frac{5e^{3x}}{9} \implies \frac{5xe^{3x}}{3} - \frac{5e^{3x}}{9} + C$$

IV Partial Fractions

For partial fractions, our goal is to guess how we “break up” the fraction, cross multiply, and then multiply by the constant we are off by. For example, consider

$$\int \frac{12}{(x^2 + 1)(x^2 + 4)}dx$$

We guess that the integral is roughly equal to

$$\int \frac{1}{x^2 + 1} - \frac{1}{x^2 + 4}dx$$

Cross multiplying gives us

$$\int \frac{3}{(x^2 + 1)(x^2 + 4)}dx$$

Note in the problem we are given 12 on the numerator, so we multiply by 4 to get it

$$4 \int \frac{1}{x^2 + 1} - \frac{1}{x^2 + 4}dx$$

Note here we split the fractions again to make the anti-derivative more clear. Integrating gives us

$$4 \left[\arctan(x) - \frac{1}{2} \arctan\left(\frac{x}{2}\right) \right] + C$$